

# QRC Usefulness Atlas: Visual Abstract

Outcome composition, family structure, anti-circularity robustness, and quantum-attribution guardrails in one reproducible view.

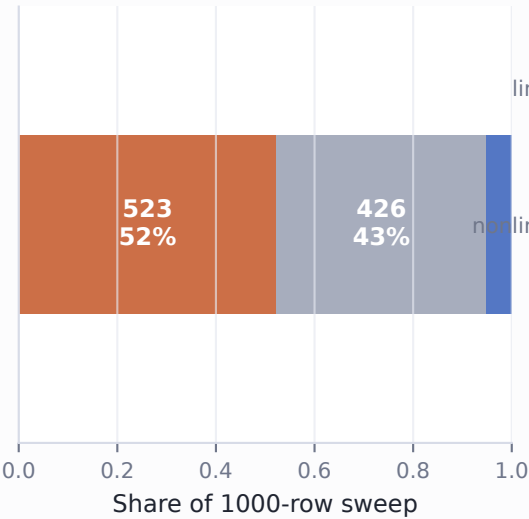
**ATLAS ROWS**  
**1,000**  
50 generators

**QRC-USEFUL**  
**51**  
5.1% of sweep

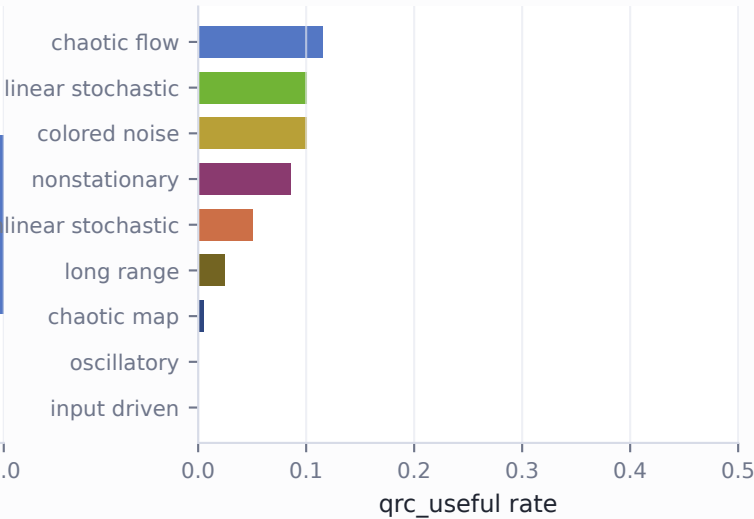
**META-MODEL**  
**R2 0.32**  
AUC 0.77

**NO-PROXY CHECK**  
**R2 0.33**  
AUC 0.75

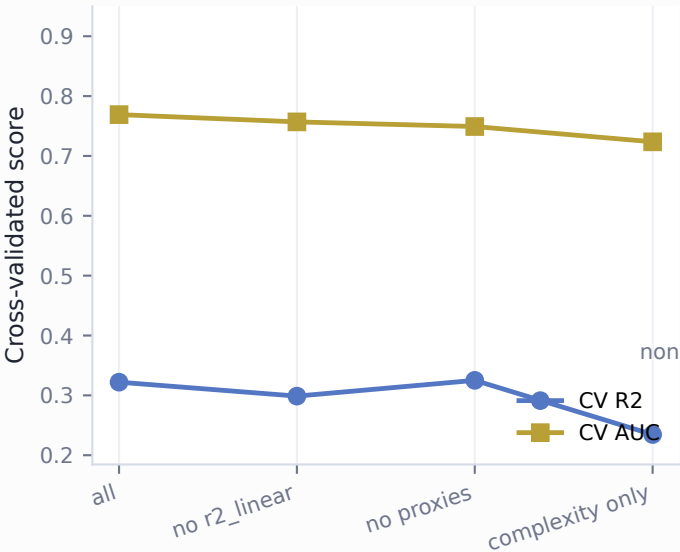
Outcome composition



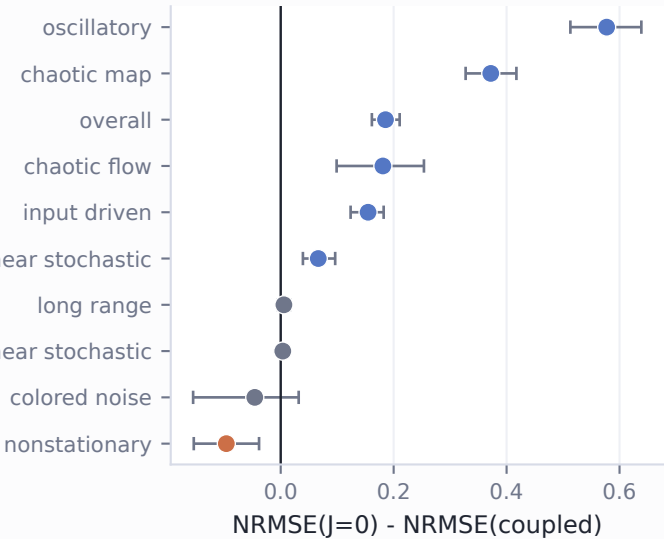
Family useful-rate ranking



Anti-circularity feature-set checks



Paired coupling-control intervals



Main claim: QRC usefulness is dataset-conditional and learnable from measured properties.

Strongest family: chaotic flow: useful rate 11.5%.

Guardrail: Coupled vs J=0 CI [0.161, 0.211], signal=positive.

Not claimed: No broad average QRC superiority and no coupling/entanglement mechanism claim.